DealMatch — System Activities and Functional Requirements (Updated)

Latest model: Category/Subcategory-based personalization. Interest-related entities and requirements have been removed.

Document status: Consolidated and synchronized with the newest DealMatch entities and foreign-key model.

1. System Overview

DealMatch is a platform for discovering, purchasing, claiming, and redeeming personalized deals and discounts. The system supports Customers, Merchants, and Administrators, with role-based permissions, deal approval, inventory management, payments, refunds, coupon/redemption processing, commissions, merchant payouts, reviews, complaints, notifications, and auditing.

2. Core System Activities

- A User registers and manages their account.
- An Administrator manages Users, Roles, and Permissions.
- A Merchant is associated with one or more Users, and Users may be associated with multiple Merchants through MerchantUser.
- An Administrator verifies and manages Merchants.
- An Administrator creates and manages Categories and Subcategories.
- A User selects one or more Categories/Subcategories to personalize Deal recommendations.
- A Merchant creates a Deal and associates it with one Category/Subcategory.
- A Merchant adds images or videos to a Deal.
- An Administrator reviews and approves or rejects Deals.
- The system publishes approved Deals when their publication conditions are satisfied.
- The system matches UserCategoryPreference records with the Category/Subcategory of eligible Deals and generates relevant recommendations or Notifications.
- A User browses Deals, searches by Category, views Deal details, and saves Deals as Favorites.
- The system checks Deal availability, eligibility, claim limits, and Inventory before allowing a User to purchase or claim a Deal.
- A User creates an Order containing one or more OrderItems when the configured purchase process requires an order.
- The system calculates SubtotalAmount, DiscountAmount, TaxAmount when applicable, and TotalAmount.
- The User makes a Payment through a supported payment provider.
- The system records Payment and PaymentTransaction information and updates the Order only after the payment result is confirmed.
- The system updates Inventory after a successful purchase and records the corresponding InventoryMovement.
- The system issues Redemption records and unique coupon codes according to the Deal's redemption rules.
- The Merchant validates and redeems the customer's coupon when the Deal is used.
- The system calculates applicable Commission for eligible sales.
- The system processes eligible Refunds and records related RefundItems.
- The system adjusts Inventory, Redemption, and Commission information according to applicable refund rules.

- The system groups eligible merchant earnings into MerchantPayouts and records related MerchantPayoutItems.
- A User can submit a Review after purchasing a Deal.
- A User can submit a Complaint regarding a Deal, Merchant, or Order.
- An Administrator can moderate Reviews and handle Complaints.
- The system records important administrative and system actions in AuditLog.

3. Roles and Functional Requirements

3.1 Customer

- Register and log in securely; recover access if credentials are forgotten.
- Select preferred Categories/Subcategories after registration and update those preferences later.
- View a personalized Deals feed and the latest Deals.
- Search Deals and filter by category, location, and price range.
- View Deal details, including original and discounted prices, discount, merchant, location, validity dates, terms, restrictions, and remaining availability where applicable.
- Add Deals to Favorites and remove them from Favorites.
- Claim or purchase an eligible Deal according to the configured business model and receive a unique coupon code and/or QR code when applicable.
- View claimed coupons, expiry dates, and claim/redemption history.
- Rate a Deal and write a Review according to platform policy.
- Receive Notifications about relevant new Deals, favorited Deals nearing expiry, and nearby Deals subject to configured preferences and location permission.
- Report a suspicious Deal or submit a Complaint.
- View order, claim, and redemption history.
- View payment and eligible refund status.
- Request a refund for an eligible purchase according to the platform's refund policy.
- View refund status and related transaction information.
- Receive Notifications about important order, payment, refund, coupon, and redemption updates.

3.2 Merchant

- Register, log in, and manage business profile information.
- Associate authorized Users with the Merchant account and manage their access according to their permissions.
- Submit required business details for verification and view verification status or rejection reasons.
- Create, update, and request deletion or cancellation of Deals.
- Upload Deal images and videos, subject to file type, size, and content validation.
- Submit Deals for Administrator approval before publication.
- Set Deal Category/Subcategory, price, discount, start/end dates, location, stock, eligibility, redemption limits, and terms.
- Manage stock and receive low-stock Alerts.
- View claims separately from completed redemptions.
- Verify coupon codes or scan QR codes and redeem only valid coupons belonging to the Merchant's own Deals.
- View redemption history and Deal performance reports.

- View Commission calculations, statuses, statements, and payout information where applicable.
- Raise a dispute about an incorrect redemption or Commission record.
- View eligible Orders and OrderItems related to the Merchant's Deals.
- Receive Notifications about claims, redemptions, low stock, Deal approval decisions, complaints, commissions, and payouts where applicable.

3.3 Administrator

- Verify, reject, suspend, and reactivate Merchant accounts; record reasons and timestamps.
- Review, approve, reject, suspend, and remove Deals; record reasons and timestamps.
- Create and manage Categories and Subcategories, including their hierarchy and active/inactive status.
- Manage customer and merchant Complaints and record their resolution.
- Monitor suspicious activity, repeated failed coupon scans, duplicate claim attempts, and unusual redemption patterns.
- View platform analytics for claims, redemptions, Deal performance, complaints, commissions, refunds, and Merchant performance.
- Manage Roles and Permissions according to least-privilege principles.
- Review AuditLog records for sensitive actions.
- Configure Commission policies and Deal approval requirements.
- Moderate Reviews.
- Manage eligible Refunds and monitor payment/refund activity.
- Investigate disputes and unusual inventory activity.
- Monitor Merchant payouts.
- Manage Notification rules and other platform-level settings.

3.4 System

- Match UserCategoryPreference with Deal.Category to support personalized Deal recommendations.
- Respect Category.ParentCategoryId when Categories are organized into Subcategories.
- Update Deal availability based on validity dates and PublicationStatus.
- Check customer eligibility, Deal availability, stock, claim limits, and redemption limits before completing a claim or purchase.
- Prevent a User from exceeding configured claim limits.
- Generate unique coupon/QR information through Redemption records.
- Validate coupon status, expiry, Deal ownership, and Merchant ownership before redemption.
- Keep claiming/purchasing separate from redemption; successful payment does not mean that a coupon has already been redeemed.
- Update Inventory and record InventoryMovement for purchases, cancellations, refunds, restocking, and manual adjustments.
- Process Payments and record payment attempts and transactions.
- Update Order and payment status only after confirmed payment-provider results.
- Process Refunds and RefundItems according to configured policy.
- Apply configured refund effects to Inventory, Redemption, and Commission records.
- Calculate and adjust Commissions when eligible sales or refunds affect Merchant earnings.

- Track MerchantPayouts and MerchantPayoutItems.
- Generate Notifications for configured events.
- Mark Notifications as read and record the read timestamp.
- Maintain AuditLog records for sensitive administrative and system actions.
- Enforce role-based permissions before protected operations.
- Protect customer, Merchant, payment, and business information.

4. Notification and Personalization Requirements

- The system shall allow Customers to select and update preferred Categories/Subcategories.
- A Category may contain Subcategories through Category.ParentCategoryId.
- The system shall associate each Deal with a Category/Subcategory through Deal.CategoryId.
- The system shall identify Customers whose UserCategoryPreference matches the Category/Subcategory of a relevant Deal.
- The system shall generate relevant Notifications or personalized Deal recommendations when matching Deals are published or become eligible according to configured rules.
- The system may notify Customers when Favorited Deals are approaching their expiry date.
- The system may notify Customers about nearby Deals when location permission is available and the relevant location feature is enabled.
- Notification records shall reference the recipient User and, when applicable, the related Deal or Order.
- Notification does not require a CategoryId because a Deal-related Notification can obtain the Deal's Category through the Deal relationship.
- The previous Interest-based model is not part of the current system: Interest, UserInterest, DealInterest, and InterestCategory are removed.

5. Order, Payment, and Refund Requirements

- The Customer shall be able to claim or purchase an eligible Deal according to the configured business model.
- The system shall create an Order when a purchase requires an order record.
- Each Order shall contain one or more OrderItems representing Deals purchased.
- The system shall calculate SubtotalAmount, DiscountAmount, TaxAmount where applicable, and TotalAmount.
- The Order shall not contain ShippingAmount because DealMatch does not require physical shipping as part of the described Deal purchase process.
- The system shall record Payment attempts and PaymentTransactions separately so failed and successful attempts can be tracked.
- The system shall not consider an Order successfully paid until the payment result has been confirmed.
- The system shall allow eligible Customers to request Refunds according to the configured refund policy.
- The system shall record the refunded OrderItems, quantities, and amounts.

6. Merchant–User Requirements

- A Merchant may be associated with one or more Users.
- A User may be associated with one or more Merchants.
- The Merchant–User relationship shall therefore be many-to-many.
- The system shall use MerchantUser to connect Merchants and Users.

- The system shall allow authorized Users associated with a Merchant to perform permitted Merchant activities according to their Roles and Permissions.
- Removing a User's association with a Merchant shall not delete the Merchant or User account.

7. Security, Audit, and Access Requirements

- The system shall require authentication before accessing protected Customer, Merchant, or administrative functions.
- The system shall use role-based access control to restrict functionality according to assigned Roles and Permissions.
- The system shall apply least-privilege access to administrative and Merchant functions.
- The system shall record sensitive administrative actions in AuditLog.
- Audit records shall identify the actor, action, affected entity, entity identifier, timestamp, and relevant details where applicable.
- The system shall prevent unauthorized Users from modifying Deals, Payments, Refunds, Commissions, Redemptions, or Merchant information.
- The system shall protect authentication credentials and sensitive payment information.

8. Reporting and Monitoring Requirements

- Customers shall be able to view their claims, coupons, Orders, Redemptions, and applicable Refunds.
- Merchants shall be able to view claim and redemption statistics for their Deals.
- Merchants shall be able to view Deal performance and Inventory information.
- Merchants shall be able to view Commission and payout information.
- Administrators shall be able to view platform-wide statistics for Deals, claims, Redemptions, complaints, Commissions, Refunds, and Merchant performance.
- Administrators shall be able to monitor suspicious activity and investigate unusual claim or redemption patterns.
- The system shall maintain sufficient historical records to support reporting, reconciliation, dispute handling, and auditing.

9. Key Updates from the Previous Version

This version is synchronized with the newest DealMatch entity model. The major change is the removal of Interest-based personalization and its replacement with Category/Subcategory-based personalization.

- **Removed:** Interest, UserInterest, DealInterest, and InterestCategory.
- **Added for personalization:** UserCategoryPreference(UserId, CategoryId, SelectedAt).
- **Category hierarchy:** Category.ParentCategoryId supports Subcategories.
- **Deal classification:** Deal.CategoryId identifies the Deal's Category/Subcategory.
- **Personalization path:** User → UserCategoryPreference → Category/Subcategory ← Deal.
- **Notification:** CategoryId remains unnecessary because Deal provides the Category relationship.
- **Order:** ShippingAmount remains removed; Order uses SubtotalAmount, DiscountAmount, optional TaxAmount, and TotalAmount.
- **Merchant–User:** remains many-to-many through MerchantUser.
- **Claim vs. redemption:** remains explicitly separated; successful payment does not automatically mean redemption.

10. Current Personalization Logic

Step 1: The Customer selects Categories and/or Subcategories that represent the types of Deals they want to receive.
Step 2: The system stores those selections in UserCategoryPreference.
Step 3: Each Deal is assigned to a Category/Subcategory through Deal.CategoryId.
Step 4: When a Deal becomes eligible for publication/recommendation, the system compares the Deal's Category with the Customer's selected Categories/Subcategories.
Step 5: Matching Customers can receive personalized recommendations or Notifications.

This approach uses the existing Category/Subcategory structure for both classification and personalization, avoiding a separate Interest layer while preserving personalized recommendations.